

BSP Release Note

for Windows[®] CE 5.0

VRender[™] 3D

Release Note

MDVR3511FBSPRN-WC50

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VR3511F

*Platform Builder for Microsoft Windows[®] CE 5.0
for VR3511F Application Processor*

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1. Release Baseline

1.1. Software Modules

Basic

Kernel

The kernel is the core of the Windows CE operating system (OS). The kernel provides the base OS functionality for any Windows CE-based device. This functionality includes process, thread, and memory management. The kernel also provides some file management functionality. The kernel services enable applications to use this core functionality. VR3511F BSP includes some kernel functions to run OS on VR3511F based system.

Features	• OS initialization code • interrupt handling of principal peripheral • Power suspend/ wake-up • Remote Kernel Debugging with KITL interface
Not Supported Features	

Ethernet Bootloader

Ethernet bootloader receives Windows CE OS image (nk.bin) from platform builder through Ethernet link and load it to RAM of the VR3511F based system. It can also store received image to the NAND flash of the system. If valid OS image is stored in NAND flash, Ethernet bootloader can load OS image from NAND flash to RAM.

Features	• Load image from Windows CE platform builder through Ethernet link • Load image from the image which is stored in NAND flash.
Not Supported Features	

Device Drivers

Display Driver

GPE Display Driver

Features	• 16 / 24 bpp • Output mode ■ LCD 480x272(WQVGA), 800x480(WVGA) • Dynamic Rotation & Output Mode Change (GUI resolution is fixed.)
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NAND Flash Storage Driver

A storage device driver that maps part of NAND flash memory as a storage device.

Feature	Can use NAND flash memory space as a storage device except of domain that stores OS image.
Limitations	- support only 8-bit bus and size of <= 128MB - The size of NAND flash memory used for storage is fixed. - One bank with 32MB(8-bit bus, 3-addr type) NAND is tested.

PCMCIA / CF Driver

A driver that facilitates access service to PCMCIA/CF memory.

Feature	Support PCMCIA / CF storage device. (CF/PCMCIA ATA storage driver that runs on this driver is required – Provided by Microsoft)
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SD/MMC Storage Driver

A storage device driver that manages read/write operation to SD/MMC memory card.

Features	- SD memory card can be mounted as a storage device whenever the system runs (hot-swap of the card is supported) - Supports SD and High-speed MMC memory card.
Limitations	Some MMC card does not work.

SERIAL Driver (Internal)

A device driver to support serial interface for VR3511F Internal serial port 1,2

Feature	Support VR3511F Internal serial port 1,2 device.
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USB 1.1 Function Driver

A driver that makes VR3511F system work as a USB device.

Feature	- USB 1.1 device controller on VR3511F is used. - VR3511F System works as USB device when a USB host is connected - ActiveSync link is possible with USB connection - Support mass storage class
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WAVEDEV Driver

Audio device driver

Feature

- support standard audio input/output interface
- Compatible with codec that use standard audio in/out

AC97 Driver

Standard AC97 Audio Driver to play wave audio files.

Feature

- Compatible with standard AC97 codec

WM9712 Driver

AC97 Audio Driver to play wave audio files and support wm9712 function

Feature

- WM9712 device controller is used.
- Compatible with standard AC97 codec
- Supports wm9712's function

SERIAL Driver (External)

A device driver to support serial interface for external serial port 1,2,3,4

Feature

- Support external serial port 1,2,3,4.

PWRBUTTON Driver

A device driver to support momentary power button switch

Feature

- Go to power suspend mode or Wake up from suspend mode, when power switch on.

SD/MMC LOADER Driver

A storage device driver loader that load SD/MMC Storage driver

Features

- Load SD/MMC Storage driver to mount SD/MMC storage card.

Limitations

- Some MMC card does not work.

TOUCH Driver

A device driver to support touch panel interface

Feature

- Support wm9712 touch controller

USB 2.0 Function Driver (Mass Storage Interface)

A driver that makes VR3511F system work as an external mass storage when connected to a USB host.

Feature	• NET2272 USB 2.0 device controller is used. • VR3511F System works as USB external storage device when USB host is connected
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WLAN Driver

A driver that facilitates communication between external wireless lan card and network protocol stack provided with Windows CE operating system.

Feature	• CF8385PN wireless lan controller is used. • Network features supported by Windows CE can be used. (Web browser, FTP server, etc.)
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Development Environment

TCP / IP Debugging

Remote debugging tools provided in development environment such as eMbedded Visual C++ or Windows CE Platform Builder can be connected to VR3511F system through TCP/IP networking.

Feature	• Supports remote debugging of applications that runs on VR3511F system through TCP/IP network (with eMbedded C++) • Remote debugging tools are available. (ex. Remote process viewer)
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KITL Debugging

Remote Kernel Debugging with KITL interface in Windows CE Platform Builder.

Feature	• Supports remote debugging of kernel or applicaton that runs on VR3511F system through KITL interface • Communicate with network device.
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2. Installation Attention

- To avoid conflict, delete previous BSP before install current BSP.
- Import new .cec file into Platform Builder's **Manage Catalog Features** after delete previous .cec.
- Refer "**BSP Build Guide For VR3511F**" document for information about making new platform and adding necessary components.

3. Known Problems

4. Release History

4.1. Version 1.0.0.beta

This is the release document of Windows CE 5.0 BSP.